Module 3 — Combinatorics

Sum rule and product rule. Permutations P(n, r) = n!/(n−r)!, combinations C(n, r) = n!/(r!(n−r)!).

Pigeonhole principle

If n + 1 objects are placed in n boxes, some box contains at least two. Generalised: some box holds at least ⌈N/k⌉ objects.

Inclusion–exclusion

|A ∪ B ∪ C| = |A| + |B| + |C| − |A∩B| − |A∩C| − |B∩C| + |A∩B∩C|. Used to count derangements: D_n = n! Σ (−1)^k / k!.

Binomial theorem

(x + y)^n = Σ C(n, k) x^k y^(n−k). Pascal's identity: C(n, k) = C(n−1, k−1) + C(n−1, k).